Test 1: Wed 9/24, in class.

Covers Transport / Laplace / Poisson Eqns.

Practice Test & Solutions in Canvas.

$$\int_U f \Delta g \, dx = -\int_U Df \cdot Dg \, dx + \int_{\partial U} f \frac{\partial g}{\partial \nu} \, dS$$

(Pf: Apply Div Thm to $\int_U \operatorname{div}(f Dg) \, dx$.)

Heat Equation

$u(x,t)$ = temperature

$$\begin{cases} u_t - \Delta u = 0 \\ u(x,0) = g(x) \end{cases}$$

$t \geq 0$, $x \in \mathbb{R}^n$ or $x \in U \subseteq \mathbb{R}^n$

$$\frac{d}{dt}\int_U u\,dx = -\int_{\partial U} F \cdot \nu \, dS$$

Diffusion $F = -aDu$

$$\Rightarrow \int_U u_t \, dx = +\int_{\partial U} aDu \cdot \nu \, dS = +a\int_U \Delta u \, dx$$

$a = 1$. Since U is anything $\Rightarrow u_t - \Delta u = 0$

$y' = -c(y - T)$

This ODE works when temp. in an object is constant in x.

Non-homogeneous heat equation:

$$\begin{cases} u_t - \Delta u = f \\ u(x,0) = g(x) \end{cases}$$

Equilibrium solutions? Yes.

Harmonic functions are equilibrium solutions:

$u(x,t) = v(x)$, $\Delta v = 0$

$\Rightarrow u_t - \Delta u = 0$

If $-\Delta u = f$, then u is equilibrium solution of $u_t - \Delta u = f$

Conservation of Energy:

Say u solves

$u_t - \Delta u = 0$ in $\mathbb{R}^n \times [0,\infty)$,

and $u(x,0) = g(x)$ is integrable on $\mathbb{R}^n$, then

$\int_{\mathbb{R}^n} u(x,t)\,dx$ is constant in time.

$$\Rightarrow \int_{\mathbb{R}^n} u(x,t)\,dx = \int_{\mathbb{R}^n} g(x)\,dx$$

for all t.

<u>Pf</u> for compactly supported u:

$$\frac{d}{dt}\int_{\mathbb{R}^n} u\,dx = \int_{\mathbb{R}^n} u_t\,dx$$

$$= \int_{\mathbb{R}^n} \Delta u\,dx$$

$$= \int_{\mathbb{R}^n} \operatorname{div}(Du)\,dx$$

$$= 0$$

Why?

$$\int_{\mathbb{R}^n} \operatorname{div}(Du)\,dx$$

$$= \lim_{R\to\infty}\int_{B(0,R)} \operatorname{div}(Du)\,dx$$

$$= \lim_{R\to\infty}\int_{\partial B(0,R)} Du\cdot\nu\,dx$$

$$= 0.$$

In general, the heat equation and Laplace's equation have analogous properties.

Look for a fundamental solution $\Phi(t,x)$.

Want: $\Phi_t - \Delta \Phi = 0$ away from $(t,x) = (0,0)$.

"$\Phi(x,0) = \delta(x)$."

$u(x,t) = \int_{\mathbb{R}^n} \Phi(t, x-y) g(y)\, dy$ will solve $\begin{cases} u_t - \Delta u = 0, \\ u(x,0) = g(x) \end{cases}$

Want Φ to respect scaling symmetries of the heat equation.

Fact: If u solves

$$u_t - \Delta u = 0,$$

so does

$$u_\lambda(x,t) = u(\lambda x, \lambda^2 t)$$

(Parabolic scaling.)

Why? $\frac{\partial}{\partial t} u_\lambda - \Delta u_\lambda = \lambda^2 u_t(\lambda x, \lambda^2 t) - \lambda^2 \Delta u(\lambda x, \lambda^2 t)$

$$= \lambda^2 (u_t - \Delta u)(\lambda x, \lambda^2 t)$$

$$= 0$$

Look for a solution $v(x,t)$ so that

$$v(x,t) = v(\lambda x, \lambda^2 t)$$

for all λ.

Choose $\lambda = \frac{1}{\sqrt{t}}$

$$\Rightarrow v(x,t) = v\left(\frac{x}{\sqrt{t}}, 1\right)$$

Let $z = \frac{x}{\sqrt{t}}$, then

$$v(x,t) = v(z,1)$$

For now, assume $n=1$.

What DE does $v(z,1) = v(z)$ satisfy in z variable?

$$V_t - V_{xx} = 0$$

$$V_t = \frac{\partial v}{\partial t} = \frac{\partial v}{\partial z}\frac{\partial z}{\partial t} = v'(z)\left(\frac{-x}{2t^{3/2}}\right)$$

$$V_x = \frac{\partial v}{\partial z}\frac{\partial z}{\partial x} = v'(z)\left(\frac{1}{\sqrt{t}}\right)$$

$$V_{xx} = \frac{1}{\sqrt{t}}\frac{\partial v'}{\partial z}\frac{\partial z}{\partial x} = \frac{1}{t}v''(z)$$

$$V_t - V_{xx} = \frac{-x}{2t^{3/2}}v'(z) - \frac{1}{t}v''(z) = 0.$$

$$z = \frac{x}{\sqrt{t}},$$

$$-\frac{x}{2t^{3/2}} v'(z) - \frac{1}{t} v''(z) = 0$$

$$-\frac{z}{2t} v'(z) - \frac{1}{t} v''(z) = 0$$

$$\longrightarrow \quad -\frac{z}{2} v'(z) - v''(z) = 0$$

ODE for $v(z)$.

Reduction of order:

$$w = v' \longrightarrow -\frac{z}{2} w - w' = 0$$

$$w' = -\frac{z}{2} w$$

$$\frac{w'}{w} = -\frac{z}{2}$$

$$\int \frac{w'}{w} dz = -\int \frac{z}{2} dz$$

$$\ln w = -\frac{z^2}{4} + c$$

$$w = c e^{-z^2/4}$$

Integrate: $v(z) = c \int_0^z e^{-s^2/4} ds$

$$v(x,t) = c\int_0^{x/\sqrt{t}} e^{-s^2/4}\,ds$$

Note: $v_t - v_{xx} = 0$

$\Rightarrow (v_x)_t - (v_x)_{xx} = 0$

$\Rightarrow v_x$ is also a solution

$$v_x = \frac{\partial v}{\partial z}\frac{\partial z}{\partial x} = c e^{-z^2/4}\cdot\frac{1}{\sqrt{t}}$$

$$= \frac{c}{\sqrt{t}} e^{-|x|^2/4t}$$

Let $\boxed{\Phi(x,t) = \frac{c}{\sqrt{t}} e^{-|x|^2/4t}}$

This is the <u>heat kernel</u>

i.e. fund. solution to

$u_t - u_{xx} = 0$.

Choose c so that

$$\int_{\mathbb{R}} \Phi(x,t)\,dx = 1$$

$$1 = \int_{\mathbb{R}} \frac{c}{\sqrt{t}} e^{-|x|^2/4t}\,dx = \frac{c}{\sqrt{t}}\int_{\mathbb{R}} e^{-|x|^2/4t}\,dx$$

$y = \frac{x}{2\sqrt{t}}$: $dx = 2\sqrt{t}\,dy$

$$\to 2c\int_{\mathbb{R}} e^{-|y|^2}\,dy = 2c\sqrt{\pi}$$

$$\to c = \frac{1}{2\sqrt{\pi}}$$

$n=1$:

$$\Phi(x,t) = \frac{1}{\sqrt{4\pi t}} e^{-|x|^2/4t}$$

$n \geq 1$:

$$\boxed{\Phi(x,t) = \frac{1}{(4\pi t)^{n/2}} e^{-|x|^2/4t}}$$

<u>Thm</u> If $g \in C(\mathbb{R}^n)$, then

$$u(x,t) = \Phi *_x g = \int_{\mathbb{R}^n} \Phi(t, x-y) g(y)\, dy$$

solves $\begin{cases} u_t - \Delta u = 0 \\ u(x,0) = g(x). \end{cases}$

<u>Remark</u>: $\Phi(t,x)$ is smooth away from $(x,t) = (0,0)$:

Note: $x \neq 0$ fixed,

$$\lim_{t \to 0^+} \frac{e^{-|x|^2/4t}}{(4\pi t)^{n/2}} = 0$$

$e^{-|x|^2/4t} \to 0$ faster than $t^{n/2}$

At $x=0$,

$$\lim_{t\to 0+} \frac{e^{-|x|^2/4t}}{(4\pi t)^{n/2}} = \lim_{t\to 0+} \frac{1}{(4\pi t)^{n/2}} = +\infty .$$

Proof of Theorem:

PDE: $u(x,t) = \int_{\mathbb{R}^n} \Phi(t, x-y) g(y)\, dy$

$$u_t - \Delta u = \int_{\mathbb{R}^n} \Phi_t(t, x-y) g(y)\, dy - \int_{\mathbb{R}^n} \Delta \Phi(t, x-y) g(y)\, dy$$

$$= \int_{\mathbb{R}^n} (\Phi_t - \Delta \Phi)(t, x-y) g(y)\, dy = 0$$

This was easy because for fixed $t>0$, $\Phi(t,x)$ is smooth in x.

Prove that $u(x,0) = g(x)$.

More precisely, $\lim_{t \to 0+} u(x,t) = g(x)$.

g is continuous, so let $x_0 \in \mathbb{R}^n$, $\varepsilon > 0$ be given, and choose $\delta > 0$ s.t.

$$|g(x) - g(x_0)| < \varepsilon \quad \text{if} \quad |x - x_0| < \delta.$$

$$u(x_0,t) - g(x_0) = u(x_0,t) - g(x_0) \int_{\mathbb{R}^n} \Phi(t, x_0 - y)\,dy$$

$$= \int_{\mathbb{R}^n} \Phi(t, x_0 - y) g(y)\,dy - \int_{\mathbb{R}^n} \Phi(t, x_0 - y) g(x_0)\,dy$$

$$= \int_{\mathbb{R}^n} \Phi(t, x_0 - y)\big(g(y) - g(x_0)\big)\,dy$$